U.S. SPACE TECHNOLOGY EXPORTS: THE CURRENT POLITICAL CLIMATE*

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1. Introduction

The U.S. climate for space technology exports has cooled considerably over the past year, and in some respects it has reached a near-freeze. All segments of the commercial space industry are feeling the impact. Two recent cases involving U.S. satellite manufacturers have received particular attention: In February 1999, the U.S. government refused to approve export licenses Hughes Electronics Corporation needed to send a communications satellite to its customer, Asia Pacific Mobile Telecommunications.[1] Loral Space and Communications Ltd. disclosed in early April 1999 that the launch of a satellite built for China would be delayed due to export controls.[2]

In addition to these high profile cases, there are numerous other incidents evidencing the chill in the space technology export climate. For example, technical discussions between U.S. and foreign companies are increasingly being hampered by export licensing delays and burdensome licensing conditions. As a result, U.S. companies are in some cases precluded from 1) sending requests for proposals ("RFP") to foreign suppliers; 2) submitting proposals to, or executing contracts with, foreign customers; or 3) sharing relevant technical information with foreign partners or even subsidiaries. In addition, some U.S. satellite companies are experiencing problems attracting foreign investors and obtaining insurance because they cannot provide pertinent technical details on the satellite system to foreign firms without a license.

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1 Warren Ferster and Sam Silverstein, *Hughes Picks Up Pieces*, SPACE NEWS Mar. 8, 1999, at 1.

2 Jeff Cole and Andy Pasztor, *Loral Says Review of Sales to China Delays New Satellite*, WALL ST. J. Apr. 2, 1999, at A3. Moreover, the preparations for the launch of six Globalstar satellites on an Ariane rocket were delayed pending approval of export licenses. Peter B. deSelding, *Export Issue Touches Europe*, SPACE NEWS Mar. 8, 1999, at 1.

The current export climate is likely to persist for some time, and as a result, U.S. companies will, in some cases, be precluded from taking advantage of business opportunities in foreign markets and foreign companies may be precluded from supplying to the U.S. market. Indeed, several companies are already placing business plans on hold and declining to pursue foreign business opportunities;[3] some U.S. companies are shying away from foreign suppliers. Companies should consider available mitigation measures, however, before taking such drastic steps. Among such measures are, for example, timely planning and preparation of technical assistance agreements ("TAA") and license applications; pre-application consultations with pertinent government agencies; careful drafting of TAAs/license requests; precise delineation of the scope of technology exchange; and appropriate follow-up. General consultations with appropriate U.S. government officials may also be helpful. Clearly, there will be situations where mitigation measures are not readily apparent or, indeed, available. In such cases, as always, a clear understanding the political dynamic driving the current export climate is critical.

2. Why the U.S. Export Climate Changed

2.1. Congressional Hearings and the Strom Thurmond Defense Authorization Act

The current chill in the export climate for space technology is a response to U.S. Congressional concerns about alleged improper technology transfers to the People's Republic of China in conjunction with launches of U.S.-made commercial communications satellites on the Chinese Long March vehicle.[4] The concerns were that U.S. companies might have transferred

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[3] For example, there have been trade press reports that U.S. satellite manufacturers declined to submit bids on a Chinese satellite project. Warren Ferster and Barbara Opall, *U.S. Firms Pass on Chinese Satellite Deal*, SPACE NEWS Nov. 23-29, 1998, at 1. U.S. companies also apparently declined to bid on a Taiwanese satellite project. Peter B. deSelding and Warren Ferster, *Dasa Wins Taiwanese Deal*, SPACE NEWS Mar. 29, 1999, at 1.

[4] Traditionally U.S. policy prohibited launches of U.S.-made satellites on Chinese launch vehicles. This policy changed in the late 1980s. In December 1988 and January 1989, the U.S. and China concluded three bilateral agreements that paved the way for launching of U.S.-made satellites on Chinese Long March vehicles. These agreements 1) established security procedures to safeguard U.S. satellite technology; 2) prohibited under-pricing of Chinese launch services and required China to price these services "on a par" with Western competitors; and 3) settled third party liability issues. Memorandum of Agreement Regarding International Trade in Commercial Launch Services, Jan. 26, 1989, U.S.-China, 28 I.L.M. 599; Memorandum of Agreement on Satellite Technology Safeguards, Dec. 17, 1988, U.S.-China, 28 I.L.M. 604; Memorandum of Agreement on Liability for Satellite Launches, Dec. 17, 1988, U.S.-China, 28 I.L.M. 609. Shortly after these agreements

improperly technical data to China during launch preparations, launch campaigns, and launch failure investigations and that such transfers, if they occurred, might have improved Chinese missile capabilities. As a result, a series of Congressional hearings and investigations followed during 1998.[5] While these hearings initially focused on the issue of improper space technology transfers to China, their scope quickly expanded to include 1) technology transfers generally to China;[6] and 2)

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were concluded, China brutally repressed pro-democracy activists in the now infamous Tiananmen Square incident, causing the U.S. Congress to pass legislation prohibiting launches of U.S. satellites on the Long March rocket, absent a Presidential waiver and Congressional notification. The *Commerce, Justice, and State, the Judiciary, and Related Agencies Appropriations Act of 1990* required a Presidential waiver and Congressional notification before a satellite can be exported to China for launch. Pub. L. No. 101-162, § 610 (1989). The *Foreign Relations Authorization Act, Fiscal Years 1990 and 1991* continued the requirement for a Presidential waiver and Congressional notification. Pub. L. No. 101-246, § 902 (1990). Waivers were granted to allow for the export of a satellite for launch on the Chinese Long March vehicle. *See* SHIRLEY A. KAN, CHINA: POSSIBLE MISSILE TECHNOLOGY TRANSFERS FROM U.S. SATELLITE EXPORT POLICY - BACKGROUND AND CHRONOLOGY, CRS REP. 98-485, at CR-12 (Aug. 13, 1998) (listing waivers). In February 1993, the U.S. and China concluded a new bilateral agreement on procedures to safeguard U.S. satellite technology. Memorandum of Agreement on Satellite Technology Safeguards, Feb. 11, 1993, U.S.-China, T.I.A.S. No. 12,486. In March 1995, the two countries renewed their Agreement on Commercial Space Launch Services. Memorandum of Agreement Regarding International Trade in Commercial Launch Services, Mar. 13, 1995, U.S.-China, 1998 BDIEL AD LEXIS 12. The new agreement allows China to launch up to eleven new satellites to geostationary orbit at prices not less than 15 percent below those charged by Western competitors for seven years. *Id.* at §§ II.B(ii), (iv). The agreement was amended in October 1997 to include terms for Chinese pricing of launch services to low Earth orbit. Agreement to Amend the Memorandum of Agreement Regarding International Trade in Commercial Launch Services, Oct. 27, 1997, U.S.-China, 1998 BDIEL AD LEXIS 10. The *Strom Thurmond National Defense Authorization Act of 1998* ("Thurmond Act") added the requirement of a Presidential certification to Congress 15 days prior to any export to China of space equipment or technology. Pub. L. No. 105-261, § 1512.

[5] Several Congressional committees asserted jurisdiction over the issues raised by the technology transfers to China. The following committees held hearings during 1998: Joint Economic Committee, Apr. 28, 1998; Senate Armed Services Committee, Jul. 9, 1998; Senate Committee on Commerce, Science, and Transportation, Sep. 17, 1998; Senate Governmental Affairs Committee, Jun. 25, 1998; Senate Governmental Affairs Subcommittee on International Security, Proliferation and Federal Services, May 21; Jun. 8; Jul. 8; Jul. 29, 1998; Senate Intelligence Committee, Jun. 4; Jun. 5; Jun. 10; Jun. 24; Jul. 8; Jul. 15, 1998; Senate Foreign Relations Committee, Jun. 11; Jun. 25, 1998; Senate Foreign Relations Subcommittee on East Asian/Pacific Affairs, Jun. 18, 1998; Joint House National Security/International Relations Committees, Jun. 17; Jun. 18; Jun. 23, 1998; House Science Committee, Jun. 25, 1998.

[6] On June 18, 1998, the House of Representatives voted 409-10 to create the Select Committee on U.S. National Security and Military/Commercial Concerns

technology transfers associated with the export of U.S.-built satellites generally and the entire licensing and enforcement process for space technology exports under the Clinton Administration.

A central issue addressed during the Congressional hearings was whether the shift in export licensing jurisdiction over commercial communications satellites from the Department of State to the Department to the Commerce ("DOC"), which was largely implemented during the Clinton Administration,[7] had unduly relaxed export controls over U.S. space technology. While the State Department applies a strict "national security" standard when evaluating export license applications, the DOC's concern for national security is tempered by a mandate to promote international trade. Critics claimed that U.S. national security had been compromised as a result of the jurisdictional shift to DOC and that stricter controls were needed for commercial communications satellites. They claimed that reduced export controls applied by DOC had provided China and other foreign countries improper access to U.S. space technology in conjunction with the foreign launches of commercial communications satellites.

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with China, chaired by Rep. Christopher Cox (R-CA) (" Cox Committee"). 144 CONG. REC. H4772 (daily ed. Jun. 18, 1998). The Cox Committee was tasked with determining whether the transfer of technology, information, advice, goods, or services to China enhanced, among other things, China's missile programs and whether China received sensitive U.S. technology. H.R. Res. 463, § 2, 105th Cong. (1998). On December 30, 1998, the Cox Committee approved a 700-page classified report on the findings of its investigation. Statement of Rep. Christopher Cox, *U.S. Representatives Discuss Technology Transfer to China*, FDCH POL. TRANSCRIPTS Dec. 30, 1998, *available in* LEXIS, News Library, Poltrn File. According to press reports, the Cox Committee found that the transfer of sensitive U.S. technology to China was "not limited to missile-satellite technology, but cover[ed] militarily significant technologies [and] that national security harm did occur." *Id.* An unclassified version of the report is expected to be released.

[7] Jurisdiction over the export of commercial communications satellites traditionally rested with the Department of State. In a series of regulatory changes throughout the 1990s, export licensing jurisdiction over these satellites was transferred to DOC. First, on October 23, 1992, export licensing jurisdiction over commercial communications satellites not containing certain specified sensitive characteristics, components, or parts were transferred to the DOC. 57 Fed. Reg. 48315 (1992). Then, on November 5, 1996, export licensing jurisdiction over commercial communications satellites incorporating these sensitive characteristics, components, or parts was transferred to the DOC, although, the sensitive characteristics, components, or parts, themselves, remained subject to State Department export jurisdiction. 61 Fed. Reg. 56894 (1996) and 61 Fed. Reg. 54540 (1996). Finally, on April 9, 1998, certain components and parts previously subject to State Department export jurisdiction (including satellite fuel, ground support equipment, test equipment, payload adapter/interface hardware, and more) were transferred to DOC jurisdiction so long as they were "to be utilized for the specific launch." 63 Fed. Reg. 17329 (1998).

The Congressional hearings culminated in the satellite export control provisions of the *Strom Thurmond National Defense Authorization Act for F.Y. '99* ("Thurmond Act"),[8] which transferred export licensing jurisdiction over commercial communications satellites back to the State Department,[9] imposed special export controls for satellites launched in certain foreign countries,[10] and introduced a requirement of certification to Congress of certain technology exports to China.[11] (See Section 4).

2.2. Implementation of the Thurmond Act and Additional ITAR Amendments

The Thurmond Act has been implemented through amendments to the International Traffic in Arms Regulations ("ITAR").[12] The ITAR amendments provide for the transfer of jurisdiction over commercial communications satellites to the Department of State and impose new requirements for technology transfer control plans and Department of Defense ("DOD") monitoring during certain technical discussions, satellite processing, launch activities, and launch failure investigations. (See Sections 4.1-4.3). Other recent ITAR amendments also impose export controls in addition to what is called for in the Thurmond Act, which have chilled further the export climate. These amendments, among other requirements, impose mandatory licensing for exports to foreign satellite insurance providers[13] and eliminate important exemptions for technical discussions with Canadian companies.[14] (See Sections 4.4-4.5).

2.3. Enforcement of the ITAR

In addition to the concrete changes in statutory law and regulations described in Sections 2.1-2.2, above, there has been a heightened sensitivity in the application and enforcement of ITAR with respect to space technology exports. The sensitivity addresses the concerns expressed by Congress with space technology exports to China and other foreign countries. Moreover, recent export licensing delays also suggest reluctance on the part of the Clinton Administration to promote space technology exports at this time, absent overriding foreign policy reasons.

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8 Pub. L. No. 105-261, §§ 1511-1516, 112 Stat. 1920, 2173-2178 (Oct. 17, 1998).

9 *Id.* § 1513.

10 *Id.* § 1514.

11 *Id.* § 1512.

12 ITAR Amendments: Control of Communications Satellites on the U.S. Munitions List, 64 Fed. Reg. 13679 (Mar. 22, 1999). *See also* Removal of Commercial Communications Satellites and Related Items from the Commerce Control List for Retransfer to the United States Munitions List, 64 Fed. Reg. 12744 (Mar. 15, 1999) (providing for the transfer from DOC).

13 64 Fed. Reg. at 13681 (codified at 22 C.F.R. § 124.15(d)).

14 ITAR Amendments, 64 Fed. Reg. 17531, 17534-35 (Apr. 12, 1999) (codified at 22 C.F.R. § 126.5).

3. The Legal Regime for Space Technology Exports

3.1. The Arms Export Control Act and ITAR

Section 38[15] of the Arms Export Control Act[16] ("AECA") provides the statutory authority for the licensing and regulation of exports of commercial space hardware and technology. The act empowers the President "to control the . . . export of defense articles and defense services and to provide foreign policy guidance to persons of the United States involved in the export . . . of such articles and services."[17] The act also provides the President authority to "designate those items which shall be considered as defense articles and defense services . . . and to promulgate regulations for the . . . export of such articles and services."[18] These designated items shall "constitute the United States Munitions List."[19]

By Executive Order, the President has delegated to the Secretary of State the authority over exports provided in the Arms Export Control Act.[20] Accordingly, the Secretary has promulgated ITAR,[21] which is the principal U.S. regulatory instrument for controlling the export of space hardware and technology.[22] The ITAR establishes the U.S. Munitions List, *i.e.*, a list of

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15 22 U.S.C. § 2778 (LEXIS 1999 through 105th Cong., 2d Sess.).

16 22 U.S.C. § 2751 *et seq.* (LEXIS 1999 through 105th Cong., 2d Sess.). One purpose of the Arms Export Control Act was to bring about a centralized and more effective control over exports within the Executive Branch and to provide a stronger voice for Congress. *International Security Assistance and Arms Export Control Act*, S. REP. No. 94-876 (May 14, 1976), at 8. U.S. export controls are designed to "further[ ] world peace and security and foreign policy of the United States." 22 U.S.C. 2778(a)(1). As for the overall purpose of the Arms Export Control Act, the act provides that "an ultimate goal of the United States continues to be a world which is free from the scourge of war and the dangers and burdens of armaments; in which the use of force has been subordinated to the rule of law; and in which international adjustments to a changing world are achieved peacefully." *Id.* § 2751.

17 *Id.* § 2778(a)(1).

18 *Id.*

19 *Id.*

20 The statutory authority to regulate exports, designate defense articles, and enforce the ITAR was delegated to the Secretary of State by Exec. Order No. 11,958, 42 Fed. Reg. 4311 (Jan. 18, 1977). *See also* 22 C.F.R. § 120.1(a) (providing that authority under Section 38 of the Arms Export Control Act was delegated to the Secretary of State).

21 22 C.F.R. Parts 120-130.

22 ITAR was first published in its present form albeit in a rudimentary version on August 26, 1955. 20 Fed. Reg. 6250 (1955). ITAR was published under the authority of the Mutual Security Act of 1954, 68 Stat. 848, which replaced the Neutrality Act of 1939, 54 Stat. 11. The provisions of the Mutual Security Act were in large measure superseded by the current Arms Export Control Act of 1976, 22

hardware and technologies deemed to require export controls in order to preserve U.S. national security.[23] The list includes *inter alia* launch vehicles, parts, and related technical data, see Category IV;[24] and satellites, parts, and related technical data, see Category XV.[25] The ITAR imposes a licensing requirement on any person who intends to export an item on the Munitions List.[26]

The ITAR is primarily administered by the State Department's Office of Defense Trade Controls ("ODTC").[27] License applications are filed with and processed by ODTC. The ODTC consults with DOD and related U.S. government defense agencies through the Defense Threat Reduction Agency ("DTRA"). The ITAR leaves broad discretion to its enforcers which infuses considerable uncertainty and variability in the application and enforcement of the regulations. United States courts have criticized ITAR for its wide latitude for discretion. For example, the district court in *Bernstein v. U.S.*[28] characterized ITAR as a "paradigm of standardless discretion."[29]

3.2. What Kind of Technical Information Requires a License to Export?

In addition to satellite and launch vehicle hardware, related technical information which amounts to "technical data" requires an export license before it can be exported. Technical data is defined to include "[i]nformation . . . which is required for the design, development, production, manufacture, assembly, operation, repair, testing, maintenance or modification of *defense articles*."[30] Technical data also includes classified information relating to defense articles and defense services;[31]

U.S.C. § 2751, *et seq.*

23 22 C.F.R. § 121.1. Certain items on the Munitions List are preceded by an asterisk, which means that the item is deemed to be "significant military equipment." *Id.* § 121.1(b). Significant Military Equipment ("SME") is defined as "articles for which special export controls are warranted because of their capacity for substantial military utility or capability." *Id.* § 120.7

24 *Id.* § 121.1, Category IV–Launch Vehicles, Guided Missiles, Ballistic Missiles, Rockets, Torpedoes, Bombs and Mines.

25 *Id.* § 121.1, Category XV–Spacecraft Systems and Associated Equipment.

26 The licensing requirement for the export of a defense article is provided in 22 C.F.R. Part 123. The licensing requirement for the export of certain defense services is provided in 22 C.F.R. Part 124. The licensing requirement for the export of unclassified technical data is provided in 22 C.F.R. Part 125.

27 22 C.F.R. § 120.1.

28 945 F. Supp. 1279 (1996), *superseded by* 974 F. Supp. 1288 (N.D. Cal. 1997), *affirmed* 1999 U.S. App. LEXIS 8595 (9th Cir. 1999).

29 945 F. Supp. at 1289.

30 22 C.F.R. § 120.10(a)(1) (emphasis added).

31 *Id.* § 120.10(a)(2). *See infra* note 45 (defining defense services).

information covered by an invention secrecy order;[32] [and] software[33] directly related to defense articles.[34]

Defense articles are, by definition, all of the items on the U.S. Munitions List.[35] Category XV on the Munitions List, entitled Spacecraft Systems and Associated Equipment, includes the following items:

> (a) Spacecraft, including [commercial[36] and military] communications satellites, remote sensing satellites, scientific satellites, research satellites, navigation satellites, experimental and multi-mission satellites;[37]
>
> (b) Ground control stations for telemetry, tracking and control of spacecraft or satellites, or employing any of the cryptographic items controlled under [the Munitions List, Category XIII].[38]
>
> (c) Global Positioning System (GPS) receiving equipment specifically designed, modified or configured for military use; or GPS receiving equipment with certain enumerated characteristics.[39]

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32 *Id.* § 120.10(a)(3).

33 Software includes but is not limited to "the system functional design, logic flow, algorithms, application programs, operating systems and support software for design, implementation, test, operation, diagnosis and repair." *Id.* § 121.8(f).

34 *Id.* § 120.10(a)(4).

35 *Id.* § 120.6.

36 *See supra* note 7 (explaining how commercial communications satellites were subject to DOC export jurisdiction from 1992 to 1998).

37 C.F.R. § 121.1, Category XV(a). Spacecraft are designated Significant Military Equipment ("SME"), which subjects them to additional restrictions under ITAR; however, commercial communications satellites, scientific satellites, research and experimental satellites are designated as SME only when the equipment is intended for use by the armed forces of any foreign country. *Id.* § 121.1, Category XV(a). The SME designation imposes requirements, e.g., with respect to submitting proposals to foreign persons, *Id.* § 126.8(a); nontransfer and use assurances, *Id.* § 124.10(a); and certifications to Congress, *Id.* § 124.11.

38 22 C.F.R. § 121.1, Category XV(b).

39 22 C.F.R. § 121.1, Category XV(c). The characteristics are as follows: 1) Designed for encryption or decryption (e.g., Y-Code) of GPS precise positioning service (PPS) signals; 2) Designed for producing navigation results above 60,000 feet altitude and at 1,000 knots velocity or greater; 3) Specifically designed or modified for use with a null steering antenna or including a null steering antenna designed to reduce or avoid jamming signals; 4) Designed or modified for use with unmanned air vehicle systems capable of delivering at least a 500 kg payload to a range of at least 300 km. *Id.* § 121.1, Category XV(c)(1)-(4). Note that "GPS receivers designed or modified for use with military unmanned air vehicle systems with less capability are considered to be specifically designed, modified or configured for military use and therefore covered under this paragraph [(c)(4)]." *Id.* § 121.1, Category XV(c).

Any GPS equipment not meeting this definition is subject to the jurisdiction of the Department of Commerce (DOC).[40]

(d) Radiation-hardened microelectronic circuits that meet certain enumerated characteristics.[41]

(e) All specifically designed or modified systems, components, parts, accessories, attachments, and associated equipment for the articles in this category.[42] [This also includes] satellite fuel, ground support equipment, test equipment, payload adapter or interface hardware, replacement parts, and non-embedded solid propellant orbit transfer engines . . . .[43]

(f) Technical data[44] [as defined above] and defense services[45] [which includes furnishing assistance, training, or technical to foreign persons] directly related to the articles enumerated in paragraphs (a) through (e).[46] This paragraph includes all technical data,

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[40] Manufacturers or exporters of equipment under DOC jurisdiction are advised that the U.S. Government does not assure the availability of the GPS P-Code for civil navigation. *Id.* It is the policy of [DOD] that GPS receivers using P-Code without clarification as to whether or not those receivers were designed or modified to use Y-Code will be presumed to be Y-Code capable and covered under this paragraph. *Id.* The DOD policy further requires that a notice be attached to all P-Code receivers presented for export. *Id.* The notice must state the following: "ADVISORY NOTICE: This receiver uses the GPS P-Code signal, which by U.S. policy, may be switched off without notice." *Id.*

[41] 22 C.F.R. § 121.1, Category XV(d). The characteristics are as follows: 1) A total dose of 5x105 Rads (SI); 2) A dose rate upset of 5x108 Rads (SI)/Sec; 3) A neutron dose of 1x1014 N/cm^2 ; 4) A single event upset of $1x10^{-7}$ or less error/bit/day; [and] 5) Single event latch-up free and having a dose rate latch-up of 5x108 Rads(SI)/sec or greater. *Id.* § 121.1, Category XV(d)(1)-(5).

[42] *Id.* § 121.1, Category XV(e).

[43] *Id.*

[44] *See supra* note 30 and accompanying text (defining "technical data").

[45] Defense service is defined as (1) [t]he furnishing of assistance (including training) to foreign persons . . . in the design, development, engineering, manufacture, production, assembly, testing, repair, maintenance, modification, operation . . . or use of defense articles; [or] (2) [t]he furnishing to foreign persons of any technical data . . . whether in the United States or abroad." 22 C.F.R. § 120.9(a).

[46] The definition was amended in 1999, and much of the text of subparagraph (f) was changed. However, the clause "as well as detailed design, development, manufacturing or production data for all spacecraft and specifically designed or modified components for all spacecraft systems" was retained. That phrase seems superfluous at best, and misleading at worst in light of the fact that *all* spacecraft are now under State Department export jurisdiction. By virtue of the definition of "technical data" this information would automatically be considered technical data. The phrase was appropriate when commercial communications satellites were under DOC export jurisdiction because even at

without exception, for all launch support activities (e.g., technical data provided to the launch provider on form, fit, function, mass, electrical, mechanical, dynamic, environmental, telemetry, safety, facility, launch pad access, and launch parameters, as well as interfaces for mating and parameters for launch).[47]

Category IV of the Munitions List concerns launch vehicles and includes *inter alia* the following items:

(a) Rockets (including but not limited to meteorological and other sounding rockets) . . . .[48]

(b) Launch vehicles and missile and anti-missile systems including but not limited to guided, tactical and strategic missiles, launchers, and systems.[49]

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(d) Missile and space launch vehicle powerplants.[50]

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that time the ITAR restricted the export of the following technical information related to commercial communications satellites: detailed design, development, production or manufacturing data for commercial communications satellites and for specifically designed or modified components for commercial communications satellites. 22 C.F.R. § 121.1 Cat. XV(g) (1998). This information was considered technical data. Note, however, the inclusion in the definition of technical data of detailed design, development, manufacturing or production information directly related to commercial communications satellites subject to DOC export jurisdiction did not include that "level of technical data . . . necessary and reasonable for a purchaser to have assurance that a U.S.-built item intended to operate in space has been designed, manufactured, and tested in conformance with specified contract requirements . . . as well as data necessary to evaluate in-orbit anomalies and to operate and maintain associated ground station equipment." *Id.*

[47] 64 Fed. Reg. 13679, 13680-13681. Technical data directly related to the manufacture or production of any article enumerated elsewhere in this category that is designated as Significant Military Equipment (SME) shall itself be designated SME. *Id.* at 13681. *See supra* note 37 (describing the significance of SME designation). Further, technical data directly related to the manufacture or production of all spacecraft, notwithstanding the nature of the intended end use (e.g., even where the hardware is not SME), is designated SME. 64 Fed. Reg. at 13681.

[48] *Id.* § 121.1, Category IV(a). These items are designated SME. *Id. See supra* note 37 (describing the significance of SME designation). In addition, this subparagraph lists "bombs, grenades, torpedoes, depth charges, land and naval mines, as well as launchers for such defense articles, and demolition blocks and blasting caps . . . ." 22 C.F.R. § 121.1, Category IV(a).

[49] *Id.* § 121.1, Category IV(b). These items are designated SME. *Id. See supra* note 37 (describing the significance of SME designation).

[50] 22 C.F.R. § 121.1, Category IV(d). These items are designated SME. *Id. See supra* note 37 (describing the significance of SME designation).

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> (h) All specifically designed or modified components, parts, accessories, attachments, and associated equipment for the articles in this category.
>
> (i) Technical data[51] [as defined above] and defense services[52] directly related to the defense articles enumerated in paragraphs (a) through (h) of this category.[53]

Technical data does *not* include "information concerning general scientific, mathematical or engineering principles commonly taught in schools, colleges and universities or information in the public domain . . . ."[54] It also does not include "basic marketing information on function or purpose or general system descriptions of defense articles."[55]

Public domain means

> information which is published and which is generally accessible or available to the public: (1) Through sales at newsstands and bookstores; (2) Through subscriptions which are available without restriction . . .; (3) Through second class mailing privileges granted by the U.S. Government; (4) At libraries open to the public or from which the public can obtain documents; (5) Through patents available at any patent office; (6) Through unlimited distribution at a conference [or] meeting [which is] generally accessible to the public, in the United States; (7) Through public release . . . after approval by the cognizant U.S. government department or agency . . .; [or] (8) Through fundamental research in science and engineering at accredited institutions of higher learning in the U.S. where the resulting information is ordinarily published and shared broadly in the scientific community . . . .[56]

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51 *See supra* note 30 and accompanying text (defining "technical data").

52 *See supra* note 45 (defining "defense service").

53 22 C.F.R. § 121.1, Category IV(i). Technical data directly related to the manufacture or production of any defense articles enumerated elsewhere in this category that are designated as SME shall itself be designated SME. *Id.*

54 *Id.* § 120.10(a)(5). *But see id.* § 124.1(a) (providing the requirement to obtain a technical assistance agreement before a defense service may be furnished even when all the information relied upon by the U.S. person in performing the defense service is in the public domain or is otherwise exempt from the licensing requirements of ITAR).

55 *Id.*

56 *Id.* § 120.11(a).

3.3. What Constitutes an Export?

The ITAR defines "export" to include 1) "[s]ending or taking a defense article out of the United States in any manner, except by mere travel by a person whose personal knowledge includes technical data . . .;" 2) "[t]ransferring registration, control or ownership to a foreign person of any . . . satellite covered by the U.S. Munitions List;" or 3) "[d]isclosing (including oral or visual disclosure) or transferring technical data to a *foreign person*, whether in the United States or abroad . . . ."[57] The first category would include, for example, sending technical data to a foreign country, e.g., by mail, express mail, facsimile, or e-mail. The third category typically would include discussion of technical data in meetings with foreign persons in the U.S. or abroad. Note that the terms "disclosure" and "transfer" could be interpreted broadly to include instances where adequate measures have not been taken to prevent access to technical data by a foreign person, e.g., where a foreign person has access to computer files or web sites containing technical data.

A "foreign person" is defined in ITAR as a foreign national,[58] except a foreign national with *permanent* resident status ("green card") in the U.S.[59] In other words, permanent U.S. residents are not considered foreign

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[57] *Id.* § 120.17(a)(4). The ITAR defines export as follows:

> (1) Sending or taking a defense article out of the United States in any manner, except by mere travel outside of the United States by a person whose personal knowledge includes technical data; or (2) Transferring registration, control or ownership to a foreign person of any aircraft, vessel, or satellite covered by the U.S. Munitions List, whether in the United States or abroad; or (3) Disclosing (including oral or visual disclosure) or transferring in the United States any defense article to an embassy, any agency or subdivision of a foreign government (e.g., diplomatic missions); or (4) *Disclosing (including oral or visual disclosure) or transferring technical data to a foreign person, whether in the United States or abroad;* or (5) Performing a defense service on behalf of, or for the benefit of, a foreign person, whether in the United States or abroad. (6) A launch vehicle or payload shall not, by reason of the launching of such vehicle, be considered an export for purposes of this subchapter. However, for certain limited purposes (see § 126.1 of this subchapter), the controls of this subchapter may apply to any sale, transfer or proposal to sell or transfer defense articles or defense services.

Id. § 120.17(a) (emphasis added).

[58] The term "national" is defined as "a person owing permanent allegiance to a state. 8 U.S.C. § 1101(a)(22) (LEXIS 1998). The term "national of the United States" means (A) a citizen of the United States, or (B) a person who, though not a citizen of the United States, owes permanent allegiance to the United States. *Id.* § 1101(a)(22).

[59] The Arms Export Control Act defines "foreign person" as "any person who is not a citizen or national of the United States or lawfully admitted to the United

persons for purposes of ITAR and may have access to technical data without a license. On the other hand, foreign nationals who reside in the U.S. on a *temporary* basis are considered foreign persons, and they may not have access to technical data without an export license. These persons include, for example, foreign students who are in the United States on an F-1 visa, exchange visitors who are in the U.S. on a J-1 visa, and foreign professionals working in the U.S. on an H-1 visa. The term "foreign person" also covers foreign corporations and other entities that are not incorporated or organized to do business in the United States, as well as international organizations, foreign governments and any agency or subdivision of foreign governments.[60]

3.4. DSP-5s and TAAs

A person who intends to export technical data needs to obtain 1) a DSP-5 license for the export of unclassified technical data;[61] or 2) a TAA for the export of defense services.[62] The ITAR defines "defense service" as "(1) [t]he furnishing of assistance (including training) to foreign persons . . . in the design, development, engineering, manufacture, production, assembly, testing, repair, maintenance, modification, operation . . . processing or use of defense articles; [or] (2) [t]he furnishing to foreign persons of . . . technical data . . . whether in the United States or abroad."[63] The DSP-5 and TAA are regulated under ITAR Parts 125 and 124, respectively, and are subject to different license application requirements, including certifications. The two licenses serve different purposes. The DSP-5 applies to the export of technical data only, and may be suitable, *e.g.*, if a company wishes to send a foreign company an RFP. The TAA contemplates a broader data exchange involving, *e.g.*, an explanation of the data or other cooperation, technology transfer, assistance or training in addition to the transfer of technical data.

States for permanent residence under the Immigration and Nationality Act, and includes foreign corporations, international organizations, and foreign governments." 22 U.S.C. § 2778(g)(9)(c) (LEXIS through 105th Cong., 2d Sess.). The ITAR definition of "foreign person" is as follows:

> any natural person who is not a lawful permanent resident as defined by 8 U.S.C. § 1101(a)(20) or who is not a protected individual as defined by 8 U.S.C. § 1324b(a)(3). It also means any foreign corporation, business association, partnership, trust, society or any other entity or group that is not incorporated or organized to do business in the United States, as well as international organizations, foreign governments and any agency or subdivision of foreign governments (e.g. diplomatic missions).

22 C.F.R. § 120.16.

60 *Id.*

61 *Id.* § 125.2(a).

62 *Id.* § 124.1.

63 *Id.* § 120.9(a)(1)-(2).

4. Recent Changes to the Legal Regime for Space Technology Exports

4.1. Transfer of Jurisdiction from Department of Commerce to State Department

Perhaps the most drastic change in the legal regime for exports of space technology is the transfer of export licensing jurisdiction over commercial communications satellites from DOC to the State Department.[64] As noted in Section 2.1, above, the jurisdictional transfer provision was a key feature of the Thurmond Act. Specifically, the act provides that "all satellites and related items . . . on the Commerce Control List of dual-use items [administered by DOC][65] shall be transferred to the United States Munitions List and controlled under [ITAR]."[66] The jurisdictional shift, which took effect on March 15, 1999,[67] was orchestrated to mitigate Congressional concern expressed in the Thurmond Act that U.S. business interests "not be placed above [U.S.] national security interests."[68]

The State Department implemented the transfer of export licensing jurisdiction in March 1999 through amendments to ITAR, and the Munitions List in particular.[69] Category XV of the Munitions List now includes all "[s]pacecraft, including [commercial and military] communications satellites, remote sensing satellites, scientific satellites, research satellites, navigation satellites, experimental and multi-mission satellites."[70] In addition, the definition of "technical data" was revised to accommodate this jurisdictional transfer. It was also broadened apparently to include certain technical information relating to launch support for foreign launches of commercial communications satellites that was not earlier considered technical data.[71] Moreover, a TAA is required before

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64 Pub. L. No 105-261, § 1513, 112 Stat. 1920, 1974 (Oct. 17, 1998). *See supra* notes 7-9 and accompanying text (discussing transfer of jurisdiction).

65 15 C.F.R. Part 774 (1998).

66 Thurmond Act, § 1513(a). *See supra* note 7 (explaining transfer to DOC).

67 *Id.* § 1513(c).

68 *Id.* § 1511(1).

69 ITAR Amendments: Control of Communications Satellites on the U.S. Munitions List, 64 Fed. Reg. 13679 (Mar. 22, 1999).

70 *Id.* at 13680 (codified at 22 C.F.R. § 121.1, Category XV(a)).

71 Included in the definition of "technical data" is all technical data

> without exception, for all launch support activities (e.g., technical data provided to the launch provider on form, fit, function, mass, electrical, mechanical, dynamic, environmental, telemetry, safety, facility, launch pad access, and launch parameters, as well as interfaces for mating and parameters for launch.)

Id. at 13680-81 (codified at 22 C.F.R. § 121.1, Category XV(f)).

defense services, presumably relating to launch support services, "may be furnished even when all the information relied upon by the U.S. person in performing the defense service is in the public domain or is otherwise exempt from the licensing requirements of [ITAR]."[72] Accordingly, these provisions elevate to the level of technical data certain form, fit and function and public domain information pertaining to launch support services not previously considered technical data.

4.2. Special Export Controls

4.2.1. The Nature of the Controls

The Thurmond Act imposes new special export controls to be applied in addition to the existing ITAR requirements for the export of a "satellite or related item."[73] These special export controls are in the form of technology transfer control plans ("TTCPs") and monitoring. The TTCPs provide in detail for the transfer of technology pursuant to a TAA or other export license and must be approved by DOD.[74] Monitoring is required for certain technical discussions, satellite processing and launch activities, and during launch failure investigations.[75]

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[72] *Id.*

[73] Thurmond Act, § 1514. Related items are defined as "satellite fuel, ground support equipment, test equipment, payload adapter or interface hardware, replacement parts, and non-embedded solid propellant orbit transfer engines . . . ." *Id.* § 1516.

[74] *Id.* § 1514(a)(1). The Thurmond Act provides that licenses for the export of these items "require a technology transfer control plan ["TTCP"] approved by the Secretary of Defense and an encryption technology transfer control plan approved by the Director of the National Security Agency. *Id.* The ITAR amendments implement this mandate, providing that "[a]ll licenses and other requests for approval require a technology transfer control plan (TTCP) approved by [DOD] and an encryption technology control plan approved by the National Security Agency." 64 Fed. Reg. 13679, 13681 (codified at 22 C.F.R. § 124.15(a)(1)). The ITAR amendments further require that applications for licenses reflect "advance discussion with [DOD]" concerning such plans. *Id.* In implementing the TTCP, there is a requirement for notification to DOD prior to any interaction with foreign persons. Thurmond Act, § 1514(a)(4); 64 Fed. Reg. 13679, 13681 (codified at 22 C.F.R. § 124.15(a)(1)).

[75] Thurmond Act, § 1514(a)(2)(A). The Thurmond Act provides for monitoring of launch activities when a license has been approved for the export of a satellite or related items for launch in certain foreign countries. The Secretary of Defense "shall monitor all aspects of the launch in order to ensure that no unauthorized transfer of technology occurs, including technical assistance and technical data." *Id.* The Thurmond Act and the ITAR require that the monitoring cover, but not be limited to–

(i) technical discussions and activities, including the design,

4.2.2. The Scope of the Special Export Controls

The ITAR imposes these controls with respect to the export "of any satellite or related item (see § 121.1. Category XV(a) and (e)) or any defense service controlled by [ITAR] associated with the launch in, or by nationals of, a country that is not a member of the North Atlantic Treaty Organization or a major non-NATO ally of the United States . . . ."[76] Moreover, ITAR provides that the special export controls "also may be imposed with respect to *any destination* as deemed appropriate in furtherance of the national security and foreign policy of the U.S."[77] This provision appears to go further than the Thurmond Act which provides that the special export controls "shall not apply to the export of a satellite or related items for launch in, or by nationals of, a country that is a member of the North Atlantic Treaty Organization or that is a major non-NATO ally of the United States."[78]

While the scope of the special export controls may seem limited to 1) the export of satellites and 2) defense services provided in connection with foreign launches primarily in non-NATO countries, the imposition of special export controls potentially reaches much farther. The ITAR provides that "the export of any article or defense service controlled under

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> development, operation, maintenance, modification, and repair of satellites, satellite components, missiles, other equipment, launch facilities, and launch vehicles;
>
> (ii) satellite processing and launch activities, including launch preparation, satellite transportation, integration of the satellite with the launch vehicle, testing and checkout prior to launch, satellite launch, and return of equipment to the United States;
>
> (iii) activities relating to launch failure, delay, or cancellation, including post-launch failure investigations; and
>
> (iv) all other aspects of the launch.

Id. § 1514(a)(2)(B)(i)-(iv); 64 Fed. Reg. 13679, 13681 (codified at 22 C.F.R. § 124.15(a)(2)(i)-(iv)). In addition, the costs of "such monitoring services shall be fully reimbursed to [DOD] by the person or entity receiving such services." Thurmond Act, § 1514(a)(2)(A). The ITAR amendments require that the U.S. licensee "make arrangements with [DOD] for monitoring," and, further, that the licensee reimburse the government for the cost of monitoring. 64 Fed. Reg. 13679, 13681 (codified at 22 C.F.R. § 124.15(a)(2)).

76 64 Fed. Reg. 13679, 13681 (codified at 22 C.F.R. § 124.15(a)).

77 64 Fed. Reg. at 13680-81 (emphasis added) (codified at 22 C.F.R. § 121.1 Category XV(f)); *see also id.* at 13681 (codified at 22 C.F.R. § 124.15(c) (providing that special export controls may be applied to the export of satellites and components for launch by members of NATO or major non-NATO allies as appropriate in furtherance of the security and foreign policy of the U.S.).

78 Thurmond Act, § 1514(b).

[ITAR] to any destination may also require that the special export controls . . . be applied."[79]

4.3. Launch Failure Investigations

The Thurmond Act provides that, in the event of a launch failure resulting from a launch of a U.S.-built satellite from a foreign country, activities in connection with a subsequent launch failure investigation are subject to ITAR regulations and licensing requirements.[80] The Thurmond Act also requires DOD monitoring of "all activities associated with the investigation [and that DOD] establish and implement a [TTCP] for the conduct of the investigation . . . ."[81] The ITAR implements these requirements,[82] while noting that such a "requirement has long existed."[83] The point made is that while there were requirements for State Department licenses in connection with the Long March launch failure investigations conducted while commercial communications satellites were subject to DOC export jurisdiction, the requirements allegedly were not followed. Such alleged failure to obtain a license under ITAR was an issue in several of the Congressional hearings that led up to the Thurmond Act.[84]

4.4. Mandatory Licenses for Insurers

Although the Thurmond Act did not address licensing requirements for the insurance community, the ITAR amendments call for mandatory licensing of exports of technical data to foreign satellite insurance providers and underwriters.[85] The regulations provide that "[n]one of the exemptions or sub-licensing provisions available in [ITAR] may be used for the export of technical data in order to obtain or satisfy insurance

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79 64 Fed. Reg. 13679, 13681 (codified at 22 C.F.R. § 124.15(c)).

80 Thurmond Act, § 1514(a)(3)(A).

81 *Id.*, § 1514(a)(3)(B)-(C).

82 The ITAR provides that the "activities of U.S. persons or entities in connection with any subsequent investigation or analysis of the failure continue to be subject to the controls established under [22 U.S.C. § 2778], including the requirements . . . for express approval prior to participation in such investigations or analyses, regardless of whether a licensed was issued . . . for the initial export of the satellite or satellite component." 64 Fed. Reg. at 13681 (codified at 22 C.F.R. § 124.15(b)(1)). In addition, the ITAR provides for DOD monitoring of "all activities associated with the investigation or analyses [and that] U.S. persons must follow the procedures" regarding a TTCP and DOD monitoring. *Id.* (codified at 22 C.F.R. § 124.15(b)(2)).

83 64 Fed. Reg. 13679, 13680.

84 *See supra* note 5 (listing Congressional hearings).

85 64 Fed. Reg. 13679, 13681 (codified at 22 C.F.R. § 124.15(d)). In addition, the regulations state that "[s]uch exports are always subject to the prior approval and re-transfer requirements of [AECA and ITAR]." *Id.*

requirements."[86] Interestingly, this licensing requirement is not new. It has always been a requirement that exports of technical data to foreign satellite insurers be licensed, just like the export to any other foreign person. Apparently, revelations that technical data pertaining to commercial satellites was sometimes being provided to foreign insurers for insurance purposes under simple non-disclosure agreements contributed to the explicit requirement for licensing.

4.5. The Canadian Exemption

Recently, the so-called Canadian exemption was revised. Previously, U.S. companies could engage in the export without a license of unclassified equipment and unclassified technical data to Canada for end use in Canada by Canadian citizens. However, on April 12, the State Department revoked that exemption with respect to, among other items, 1) launch vehicles and rockets, 2) "Spacecraft, Remote Sensing Satellites, and Military Communications Satellites . . ." and 3) technical data related to those items.[87] Note that commercial satellites are not explicitly included in the list of items for which the exemption is revoked. On the other hand, they are "spacecraft," which are explicitly covered. Is the intention that they continue to be subject to the Canadian exemption? What about launch vehicle and satellite parts? Are they still covered under the Canadian exemption? Parts are not included among the items for which the Canadian exemption was revoked.[88] However, if the parts are on the Missile Technology Control Annex,[89] they are subject to the exemption.[90] Note also that the Canadian exemption does not apply to "[a]ny defense service covered by part 124 . . . ."[91] The revocation of the Canadian exemption occurred after bilateral negotiations between the U.S. and Canada over the past year. The negotiations were prompted by a concern on the part of U.S. authorities about possible unauthorized end-use of items exported to Canada pursuant to the exemption.

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86 *Id.*

87 ITAR Amendments, 64 Fed. Reg. 17531, 17534 (Apr. 12, 1999) (codified at 22 C.F.R. § 126.5(b)). The ITAR provides that the "exemption . . . does not apply to the following [defense articles, defense services and related technical data]: Category IV (a), (b), (c), (d), (f), and (g)–Launch Vehicles, Guided Missiles, Ballistic Missiles and Rockets [and] Spacecraft, Remote Sensing Satellites, and Military Communications Satellites listed in Category XV(a), (b), and (c) . . . ." *Id.*

88 *See id.* (codified at 22 C.F.R. §§ 126.5(b)(3) and (9)) (omitting Category IV(h) and Category XV(e) which include parts and components).

89 Missile Technology Control Regime, 32 I.L.M. 1298 (Sep. 1993). The MTCR is an arrangement between 29 countries to adhere to a common set guidelines for restricting the export of missile equipment and technology.

90 64 Fed. Reg. 17531, 17534 (codified at 22 C.F.R. § 126.5(b)(13)).

91 *Id.* (codified at 22 C.F.R. § 126.5(d)).

Conclusion

There is no doubt that the U.S. export climate for space technology has cooled significantly and all segments of the space industry are feeling the impact. It is likely that the current U.S. export climate will persist for some time – at least until the 2000 U.S. Presidential elections. The Thurmond Act, the ITAR amendments, and heightened political sensitivity to space technology exports will continue to influence the application and enforcement of U.S. export controls. Now, more than ever, carefully designed and implemented mitigation measures are critical to overcome real and perceived obstacles, as is a clear understanding of the underlying political dynamic driving the export licensing environment.